Shirpur Education Society's

R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR

An Autonomous Institute

(Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere)

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(स्वायत्त महाविद्यालय)

Programme: B.TECH (AIML)/B.TECH (COMP)/B.TECH (DATA SCI.)

Year: II/Semester III (Exam Year: 2023-2024)

Subject: Database Management Systems **Max Marks**: 65

Date: 10 Jan 2024 **Time:** 02:30 pm - 04:30 pm (02:00 Hrs.)

END SEMEATER EXAMINATION ODD SEM III (Acad. Year: **2023-2024**)

Instructions:

1. This question paper contains 3 pages
2. All Questions are Compulsory.
3. Answer to each new question is to be started on a fresh page.
4. Assume suitable data wherever required, but justify it.
5. Draw the neat labelled diagrams, wherever necessary.
6. Figures on the right indicate full marks.

1. Q1 **20**

A. **10**

i. Construct an E-R (Entity Relationship) diagram for hospital with a set of patients and set of doctors. Patients are treated in a single ward by doctors assigned to them. Each patient will be treated by a single doctor. Healthcare assistants also attend to the patients, some of these are associated with each ward. Patients undergo various tests. The accounts department manages patient's treatment bill and staff payment. Some staff are paid part-time and doctors, and healthcare assistants work at varying rates (subject to grade). Assume appropriate attributes wherever required. **10**

---------- OR ----------

ii. Suppose you are given a relation R= (A, B, C, D, E) with the following functional dependencies: **10**
{BC → ADE, D → B}
a. Compute candidate keys.
b. Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF).

c. If the relation is not in BCNF, decompose it until it becomes BCNF.
At each step, identify anew relation, decompose and re-compute the keys and the normal forms they satisfy

B. **10**

i. What is a deadloack? Discuss the methods used for deadlock prevention? **5**

ii. Explain Atomicity and Consistency properties of transaction. **5**

2. Q2 **15**

A. Apply timestamp based protocol on the following schedule and determine whether it will lead to successful execution or not. Describe the steps taken to execute read and write conflicting operations. Also determine the final values of read and write timestamps of item A and B. **10**

T1	T2
BEGIN	
R(B)	
	BEGIN
	R(B)
	W(B)
R(A)	
	R(A)
	W(A)
	COMMIT
COMMIT	

B. **5**

i. What do you mean Database Schema and Instance. Discuss with the help of examples. **5**

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ii. Discuss the characteristics of the Database System. **5**

3. Q3 **15**

A. Consider the following database schema: **10**
Employee(employee_name, street, city, date_of join)
Works(employee_name ,company_name, salary)
Company(company_name, city)
Manages(employee_name, manager_name) |
Solve the following queries using SQL:
1.Give all employee of ABC Company a 25% rise(2M).
2. Find all employees who live in the same cities and on the same street as their manager(2M).

3. Find all employees who join in the month of April(1M).
4. Delete the employee Jennifer belonging to XYZ Company(2M).
5. Find the minimum and maximum salary for each company, ordered by company name(1M).
6. Find the average salary for each city, and list the cities in alphabetical order(2M).(2M).

B. **5**

i. Discuss how privileges are granted and revoked. **5**

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ii. Explain the significance of authorization in SQL databases. **5**

4. 4 **15**

A. **10**

i. Describe following basic operators in Relational Algebra with an example **10**
SELECT
PROJECT
UNION
SET DIFFERENCE
CARTESIAN PRODUCT

---------- OR ----------

ii. Discuss the disjoint and overlap constraints in specialization. **10**

B. **5**

i. Discuss single level indexes in brief. **5**

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ii. Discuss Static Hashing Techniques in detail. **5**